

PigeOff – Evaluation

The following document reports the evaluation of the PigeOff system, the final project for the Internet-of-Things course at Sapienza University.

The contributors are listed below:

- Anja Škrlić - 2285543
- Filippo Zanei - 2285059
- Teun Boekholt - 2284223

while the GitHub repository is available at: <https://github.com/TeunBoekholt/Pigeoff>

The evaluation was mainly done with respect to the two most important constraints of the project: the system has to react before the pigeon leaves the monument, and it has to be able to operate for at least one week on battery power. In addition to this, we also evaluated the reliability of the LiDAR trigger and of the ESPCAM classifier at different distances.

Field observation

Before evaluating the system in a controlled environment, we observed pigeon activity directly on monuments in Rome. During one observation session at a fountain statue in Piazza Vittorio Emanuele, 20 pigeons and 6 other birds landed on the statue in one hour, between 11:00 and 12:00. The average permanence time of the pigeons on the statue was around 16.3 seconds.

This result is important because it gives a realistic upper bound for the detection and reaction time of the system. If the system takes longer than this, it risks detecting the pigeon too late. For this reason, in the design phase we considered 10 seconds as the maximum acceptable duration for the full wake-up, detection, classification and response routine.

On another monument, instead, no pigeons landed during one hour of observation, even though several pigeons were present nearby. This suggests that pigeon activity is strongly dependent on the specific place and on the time of day. Therefore, the field observation should be considered useful for defining the timing constraint, but not sufficient to fully characterize pigeon behaviour in Rome.

Detection performance

The detection performance was evaluated in a controlled setup by testing the system at three different distances: 20 cm, 40 cm and 80 cm. The system was tested with three different conditions: a pigeon image, a branch, and an empty white background used to simulate the surface of a statue.

The results are reported in the following table.

Distance	LiDAR FN	LiDAR FP	Camera FN	Camera FP	LiDAR F1	Camera F1
20 cm	0%	20%	10%	0%	88.89%	94.74%
40 cm	10%	0%	40%	0%	94.74%	75.00%
80 cm	30%	0%	60%	0%	82.35%	57.14%

From these results we can say that the LiDAR alone performs best at 40 cm, where it reaches an F1 score of 94.74%. However, the complete system also depends on the ESPCAM classifier, and the camera performs much better at 20 cm. At this distance, the camera reaches an F1 score of 94.74%, with only 10% false negatives and no false positives.

For this reason, considering the full LiDAR + ESPCAM system, the best placement distance is around 20 cm. At this distance the LiDAR has a higher false positive rate, probably because it can also detect branches or similar objects, but this is partially compensated by the camera classifier, which is used as the second validation step.

At 80 cm the system is not reliable enough. The LiDAR false negative rate increases to 30%, and the camera false negative rate reaches 60%. Therefore, this distance should not be used for the current prototype.

Timing evaluation

The average time required by the system after a trigger is around 9 seconds. This includes the activation of the ESPCAM, the camera stabilization time, the image capture, the local inference and the response signal back to the ESP32.

Since the average permanence time of a pigeon observed on the statue was 16.3 seconds, the system is fast enough to detect the pigeon while it is still present. From this point of view, the prototype satisfies the timing constraint.

However, this result was obtained in a controlled setup. In a real outdoor installation, lighting conditions, camera exposure, shadows and sensor noise could increase the time needed to obtain a reliable classification. For this reason, the 9 seconds result is promising, but should be validated with longer outdoor tests.

Energy consumption

In order to evaluate the possibility of reaching one week of battery life, we measured the energy consumption of the main parts of the prototype using an INA219 sensor. The ultrasound actuator was not implemented in the prototype, so it was not included in the energy calculation.

In the original idea, the system should have used an LD2410S radar sensor as the always-on trigger. Since the radar sensor was not available during the project, we used a TF-Mini LiDAR instead. The LiDAR allowed us to test the wake-up routine and the general behaviour of the system, but it is not a good solution for the final product because its energy consumption is too high.

For the ESPCAM, we measured an average sleeping power of 26 mW. A detection with pigeon recognition consumed around 0.34 mWh, while a detection without pigeon recognition consumed around 0.23 mWh. This confirms that the ESPCAM can be used in an event-based way, as long as it stays in deep sleep for most of the time.

For the LiDAR, instead, the measured power consumption was much higher. For distances smaller than 40 cm it consumed around 245 mW, at 80 cm around 283 mW, and outdoors, when measuring distances above 300 cm, it consumed more than 450 mW. This makes it unsuitable as an always-on trigger for a battery-powered deployment.

The Heltec ESP32 v3 was also measured during operation. In light sleep mode it consumed around 46 mW, while in active mode it consumed around 204–220 mW. During LoRa transmissions, the power peaks were higher than 800 mW. From the measurements, we estimated that one full routine after a trigger takes around 9 seconds and consumes around 0.54 mWh.

For the final battery estimation we therefore considered the theoretical radar sensor instead of the LiDAR. From the nominal values of the LD2410S radar sensor, with a voltage of 3.3 V and a typical current of 0.12 mA, the theoretical power consumption is: 0.396mW

This is much lower than the measured consumption of the LiDAR, and it is the main reason why the final system should use the radar sensor.

Battery life estimation

To estimate the battery capacity required for one week of operation, we considered a conservative scenario. We assumed 16 activations per hour, corresponding to the peak activity observed during the field test. We also assumed that the system operates continuously during day and night, that the radar sensor is always active, and that the battery provides 5 V.

Under these assumptions, the estimated hourly consumption is:

Component	Estimated consumption per hour
ESPCAM	34.3 mWh
Radar sensor	0.4 mWh
Heltec ESP32 v3	134.8 mWh
Total	169 mWh

Starting from the total hourly consumption, the weekly energy requirement is:

$$169(mWh) \cdot 24 \cdot 7 = 28472(mWh)$$

Assuming a 5 V battery, the corresponding battery capacity is:

$$28472(mWh) / 5V = 5694mAh,$$

Therefore, under the assumptions used in this evaluation, the system could theoretically work for one week using a battery with a capacity comparable to a large smartphone battery. This result suggests that the one-week autonomy requirement is realistic, provided that the final system uses the low-power radar sensor instead of the LiDAR.

Discussion

From the evaluation we can say that the prototype is able to satisfy the two main requirements of the project. The detection routine takes around 9 seconds, which is lower than the average observed pigeon stay time of 16.3 seconds. The energy estimation also shows that one week of operation is possible with a realistic battery size.

At the same time, the prototype is not yet ready for a real deployment. The main weakness is the ESPCAM classifier. The best results were obtained at 20 cm and in a high-contrast situation, where the pigeon is dark and the background is light. This condition is reasonable for some monuments, but it cannot be guaranteed in every real environment. Shadows, different statue materials, dirt, other birds and changing sunlight could reduce the reliability of the classifier.

Another limitation is that the dataset used for the final classifier was very small. Even though data augmentation was used, the model should be trained and tested with more images taken directly from the ESPCAM in realistic outdoor conditions. This would make the evaluation more representative of the final deployment.

Finally, the ultrasound actuator was not implemented, so its energy consumption and real deterrent effect were not evaluated. Also, the radar sensor was only considered theoretically, because the final sensor was not available during the project. For a future version, both the radar and the actuator should be tested directly as part of the complete system.

Conclusion

In conclusion, the evaluation confirms that PigeOff is a feasible proof of concept. The prototype is able to detect a pigeon-sized target, activate the ESPCAM, run local inference and respond within the required time window. Moreover, the estimated energy consumption shows that a one-week battery life is achievable if the final version uses the intended low-power radar sensor.

The current system is therefore promising, but its reliability is still limited by the camera classifier and by the controlled nature of the tests. The next step should be a longer outdoor deployment, using the final radar sensor and the ultrasound actuator, in order to measure the real battery life and the real deterrent effect of the system.